1 Module 1: basic topological structures and invariants

1.1 Metric spaces

A ***metric space*** is a pair (X, d) consisting of a set X and a function $d\colon X \to R_{\geq 0}$, called the ***metric***, that satisfies the following three properties:

- $d(x, x) = 0$ for every $x \in X$,
- $d(x, y) > 0$ for every $x \neq y$ (*positivity*),
- $d(x, y) = d(y, x)$ for any $x, y \in X$ (*symmetry*),
- $d(x, z) \leq d(x, y) + d(y, z)$ for any $x, y, z \in X$ (*triangle inequality*).

An ***open (resp. closed) ball*** of radius $r > 0$ with center x_0 in a metric space (X, d) is the set $B(x_0, r) = \{x \in X \mid d(x_0, x) < r\}$ (resp. $\bar{B}(x_0, r) = \{x \in X \mid d(x_0, x) \leq r\}$).

1.1.1 Example (the real line R)

The real numbers R with the absolute value metric $d = |\cdot|$ constitute a metric space:

- $d(x, x) = |x - x| = 0$,
- $d(x, y) = |x - y| > 0$ for $x \neq y$,
- $d(x, y) = |x - y| = |y - x| = d(y, x)$,
- $d(x, z) = |x - z| = |(x - y) + (y - z)| \leq |x - y| + |y - z| = d(x, y) + d(y, z)$.

The balls are open and closed intervals:

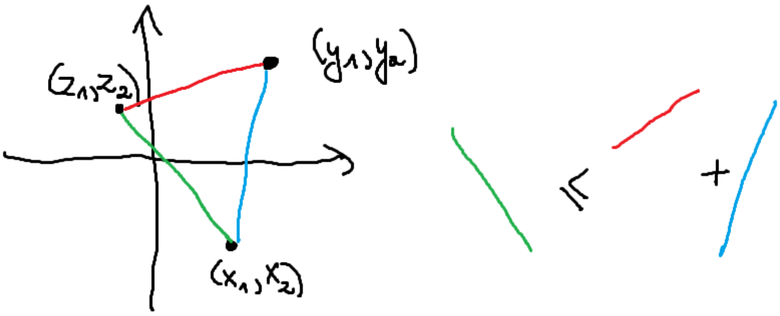
1.1.2 Example (the plane R^2)

The plane R^2 equipped with the Euclidean metric

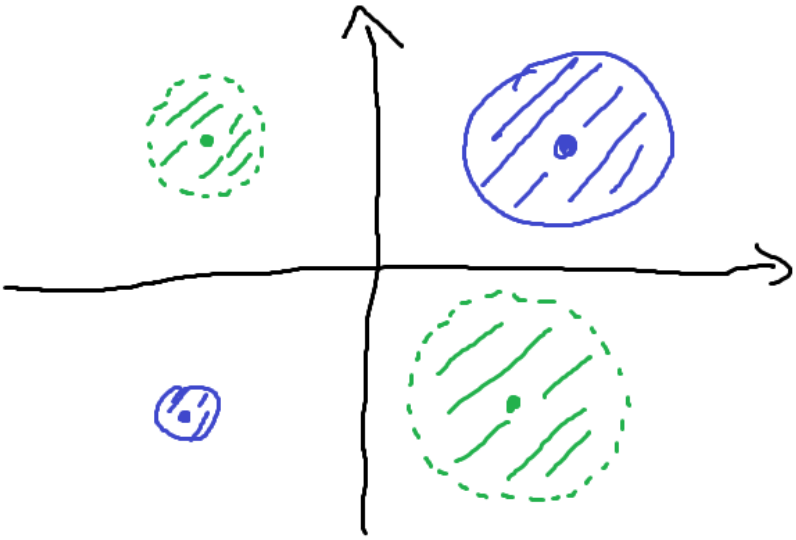
$$d((x_1, x_2), (y_1, y_2)) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

forms a metric space:

- $d((x_1, x_2), (x_1, x_2)) = \sqrt{(x_1 - x_1)^2 + (x_2 - x_2)^2} = 0$,
- $$\begin{aligned} d((x_1, x_2), (y_1, y_2)) &= \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2} \\ &= \sqrt{(y_1 - x_1)^2 + (y_2 - x_2)^2} \\ &= d((y_1, y_2), (x_1, x_2)). \end{aligned}$$
- the triangle inequality in this case can be visualized in the picture below — it corresponds directly to the familiar geometric triangle inequality.



The balls are open and closed disks:



1.1.3 Example (R^n)

R^n is also equipped with the Euclidean metric

$$d((x_1, \ldots, x_n), (y_1, \ldots, y_n)) = \sqrt{(x_1 - y_1)^2 + \ldots + (x_n - y_n)^2}$$

- $d((x_1, \ldots, x_n), (x_1, \ldots, x_n)) = \sqrt{(x_1 - x_1)^2 + \ldots + (x_n - x_n)^2} = 0,$
- $$\begin{aligned} d((x_1, \ldots, x_n), (y_1, \ldots, y_n)) &= \sqrt{(x_1 - y_1)^2 + \ldots + (x_n - y_n)^2} \\ &= \sqrt{(y_1 - x_1)^2 + \ldots + (y_n - x_n)^2} \\ &= d((y_1, \ldots, y_n), (x_1, \ldots, x_n)). \end{aligned}$$

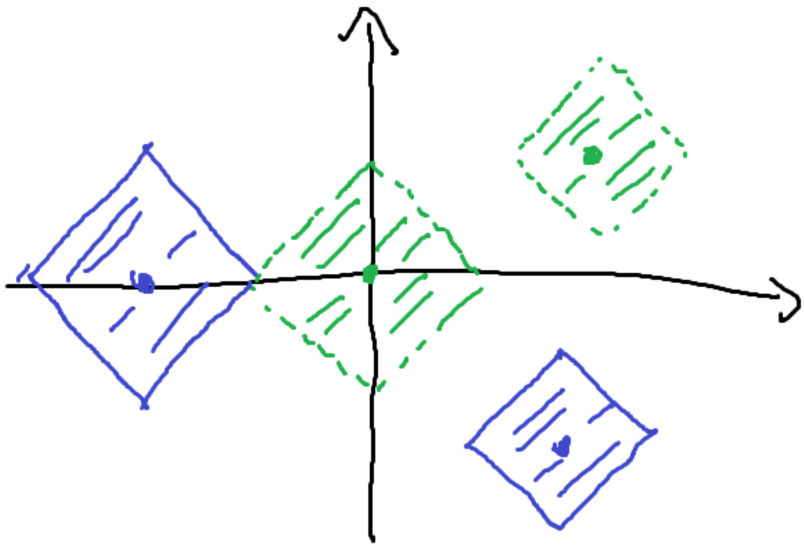
To prove the triangle inequality, we recall the *Cauchy–Schwarz* inequality: $(u_1 v_1 + \ldots + u_n v_n) \leq \sqrt{u_1^2 + \ldots + u_n^2} \cdot \sqrt{v_1^2 + \ldots + v_n^2}$. Then the proof proceeds as follows:

•
$$\begin{aligned}
d((x_1,\ldots,x_n),(z_1,\ldots,z_n))^2 &= (x_1-y_1+y_1-z_1)^2+\ldots+(x_n-y_n+y_n-z_n)^2\\
&= \sum_{i=1}^n (x_i-y_i)^2+\sum_{i=1}^n (y_i-z_i)^2+2\sum_{i=1}^n (x_i-y_i)(y_i-z_i)\\
&\leq \sum_{i=1}^n (x_i-y_i)^2+\sum_{i=1}^n (y_i-z_i)^2+2\sqrt{\sum_{i=1}^n (x_i-y_i)^2}\cdot\sqrt{\sum_{i=1}^n (y_i-z_i)^2}\\
&= \left(\sqrt{\sum_{i=1}^n (x_i-y_i)^2}+\sqrt{\sum_{i=1}^n (y_i-z_i)^2}\right)^2\\
&= \left(d((x_1,\ldots,x_n),(y_1,\ldots,y_n))+d((y_1,\ldots,y_n),(z_1,\ldots,z_n))\right)^2.
\end{aligned}$$

1.1.4 Example (Manhattan metric)

(R^2,d), $d((x_1,x_2),(y_1,y_2))=|x_1-y_1|+|x_2-y_2|$.

The balls look as follows:



1.1.5 Example (discrete metric)

(X,d),

$$d(x,y)=\begin{cases}0 & x=y\\ 1 & x\neq y\end{cases} \tag{1}$$

The balls simplify significantly in this metric: there are two types of open balls — those consisting of single points (for radius ≤ 1) and the one consisting of the entire space X (for any radius greater than 1).

1.1.6 Example (a metric on the Cartesian product)

Suppose (X,d_X) and (Y,d_Y) are metric spaces. Then, the Cartesian product $X\times Y$ can be equipped, for example, with the following metric:

$$d((x_1,y_1),(x_2,y_2))=d_X(x_1,x_2)+d_Y(y_1,y_2). \tag{2}$$

Note that if $X=Y=R$ and $d_X=d_Y$ is the absolute value metric, then this Cartesian product metric coincides with the Manhattan metric.

1.1.7 Open subsets in metric spaces

A subset $A \subseteq X$ of a metric space (X, d) is ***open*** if, for every $x \in A$, there exists $\varepsilon > 0$ such that $B(x, \varepsilon) \subseteq A$.

Example The empty set $\emptyset$ and the entire metric space X are examples of open subsets of a metric space (X, d).

Example Unions and finite intersections of open subsets are also open subsets.

1.2 Topological spaces

A ***topological space*** is a pair (X, τ), where τ is a collection of subsets of X, called the ***topology*** of X, satisfying the following conditions:

- $\emptyset, X \in \tau$,
- any union (finite or infinite) of subsets from τ belongs to τ,
- finite intersections of subsets from τ belong to τ.

The subsets in τ are called ***open*** subsets of X.

Note that every metric space naturally induces a topological space — see the last two examples.

1.2.1 Example (trivial and discrete topologies)

Trivial topology: $\tau = \{\emptyset, X\}$

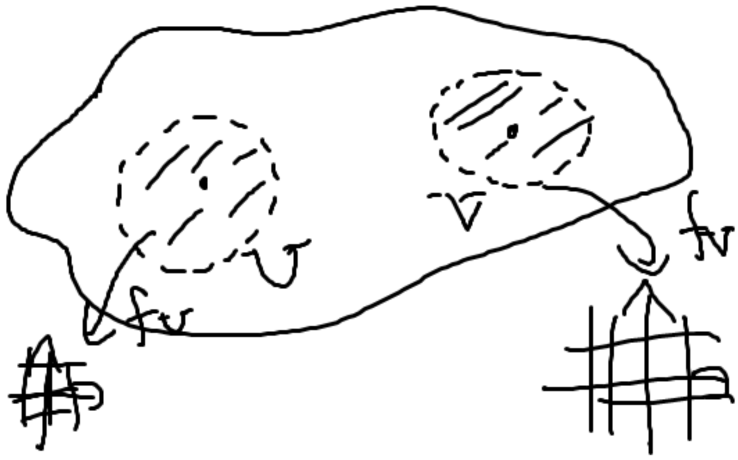
Discrete topology: $\tau = \{A \mid A \subseteq X\}$. Note that the discrete metric on X induces precisely the discrete topology.

1.2.2 Example (manifolds)

Before we give the definition of a manifold, let us recall the necessary concepts:

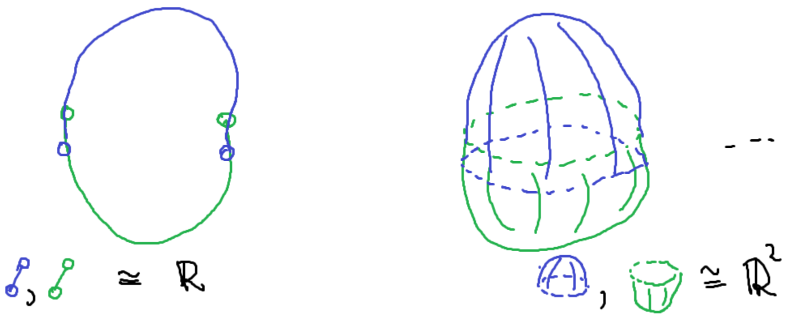
- a function $f\colon (X, \tau_X) \to (Y, \tau_Y)$ is ***continuous*** if, for every open subset $V \subseteq Y$ (i.e., $V \in \tau_Y$), its preimage $f^{-1}(V)$ is open in X (i.e., $f^{-1}(V) \in \tau_X$),
- a ***homeomorphism*** is a continuous bijection $f\colon X \to Y$ whose inverse is also continuous,
- a topological space X is ***Hausdorff*** if any two distinct points have disjoint open sets containing them.

A ***topological manifold*** of ***dimension*** n is a Hausdorff topological space M together with a collection of homeomorphisms $f_U\colon U \to R^n$, where the U's are open subsets of M that cover M. The collection $\{U, f_U\}$ is called the ***atlas*** of M.

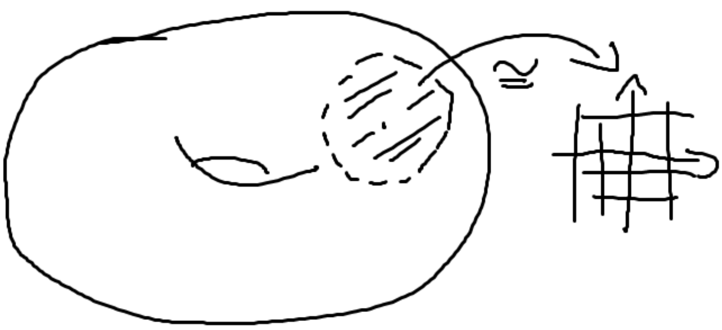


Example (R^n) This is probably the simplest example of a manifold — the atlas can be chosen to consist of a single chart (i.e., an open subset together with the corresponding homeomorphism), which is the entire space with the identity homeomorphism.

Example (spheres)

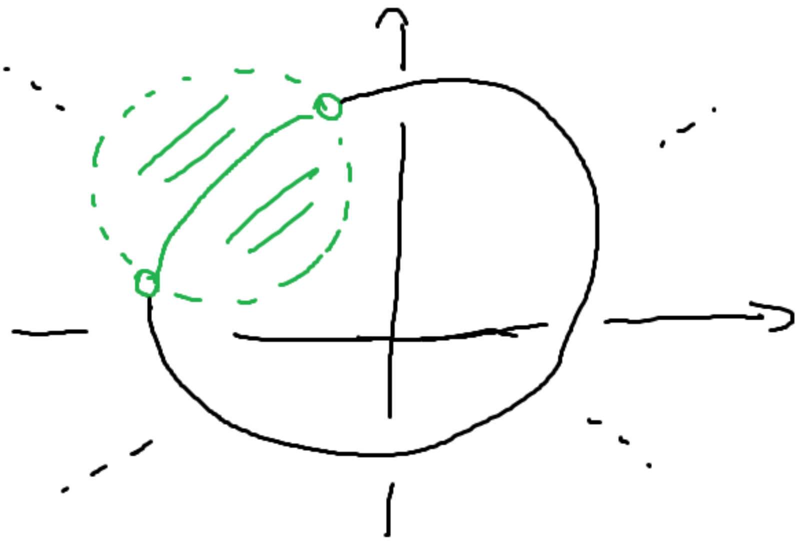


Example (torus)



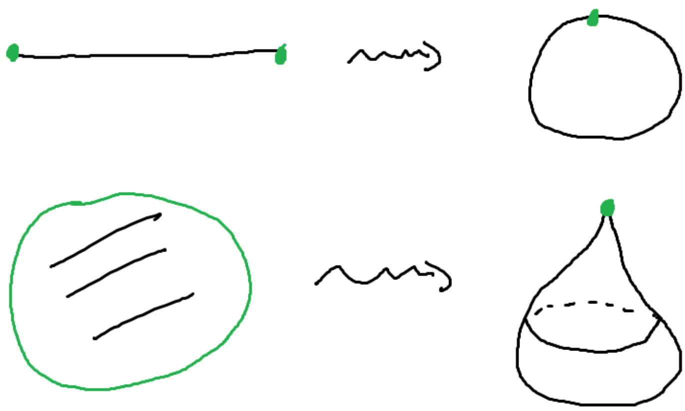
1.2.3 Example (subspace topology)

Given a subset A of a topological space (X, τ_X), we define the ***subspace*** topology τ_A on A as follows: $\tau_A = \{A \cap U \mid U \in \tau_X\}$.



1.2.4 Example (quotient topology)

Given an equivalence relation $\sim$ on a topological space X, the quotient topological space of X with respect to $\sim$ is $([X]_\sim, \tau_{[X]_\sim})$. We define the ***quotient*** topology $\tau_{[X]_\sim}$ on it by letting the open sets (i.e., those belonging to $\tau_{[X]_\sim}$) be precisely the subsets U of $[X]$ such that the set of $x \in X$ with $[x] \in U$ is open in X (i.e., belongs to τ_X).



1.2.5 Alternative definition of a topological space (via closed subsets)

A ***topological space*** is a pair (X, τ), where τ is a collection of subsets of X, called the ***topology*** of X, satisfying the following conditions:

- $\emptyset, X \in \tau$,
- any intersection (finite or infinite) of subsets from τ belongs to τ
- finite unions of subsets from τ belong to τ.

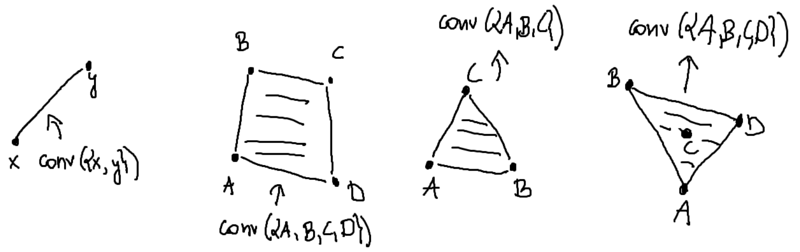
We call the subsets from τ ***closed*** subsets of X.

The two definitions of a topological space presented in these notes are equivalent, and the closed subsets are precisely the complements of the open subsets.

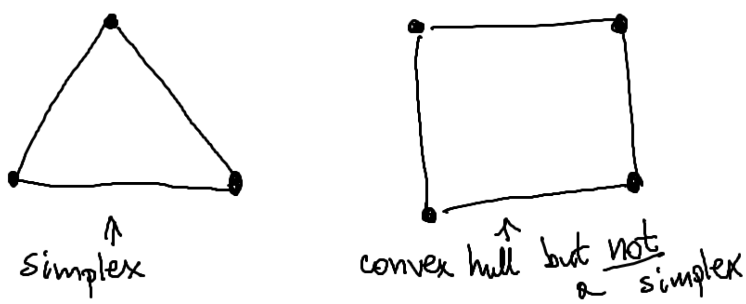
1.3 Simplicial complexes

1.3.1 Definition of a simplex

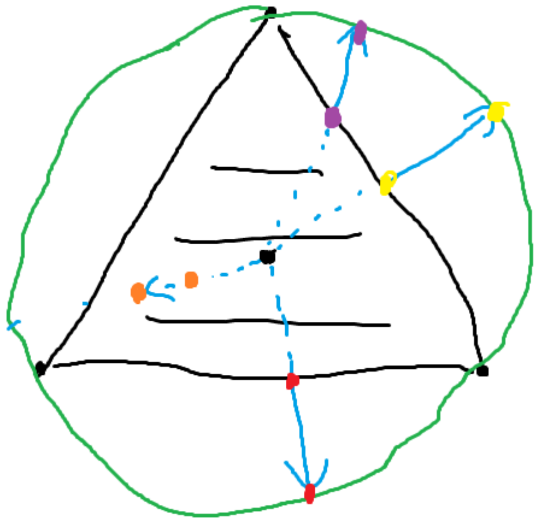
A set $X \subseteq R^d$ is called ***convex*** if, for any two points $x, y \in X$, the line segment $[x, y] = \{tx + (1-t)y : t \in [0,1]\}$ is contained in X. The ***convex hull*** of a set $X \subseteq R^d$ is the smallest convex set $\mathrm{conv}(X)$ containing X.



The convex hull of a set of $n+1$ points $\{a_0, a_1, \ldots, a_n\} \subset R^d$, such that the vectors $a_1 - a_0, \ldots, a_n - a_0$ are linearly independent, is called a (geometric) n***-dimensional simplex*** in R^d. We denote it by $\Delta[a_0, \ldots, a_n]$ or $[a_0, \ldots, a_n]$. The points $a_0, \ldots, a_n$ are called the ***vertices*** of the simplex.



The n-simplex Δ^n is homeomorphic to the n-dimensional disc $D^n = \{x \in R^n : ||x|| \leq 1\}$:



1.3.2 Definition of a simplicial complex

A ***simplicial complex*** is a set K of simplices in R^d satisfying the following conditions:

- if a simplex belongs to K, then each of its *faces* (that is, simplices formed by proper subsets of the vertices of the considered simplex) also belongs to K,
- the intersection of any two simplices σ and τ from K is either a face of both σ and τ,
- (*local finiteness condition*) each vertex of a simplex from K is contained in finitely many simplices from K.

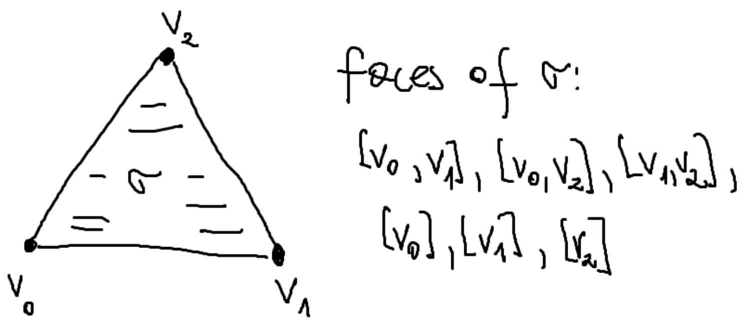
Let $|K| \subseteq R^d$ be the union of the simplices of K:

$$|K| = \bigcup_{\sigma \in K} \sigma. \tag{3}$$

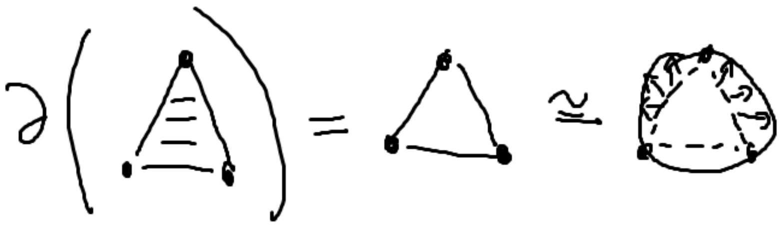
We define the topology of $|K|$ by declaring a subset $A \subseteq |K|$ to be closed if and only if $A \cap \sigma$ is closed in σ for each $\sigma \in K$, where each simplex has its natural topology as a subset of R^d. The space $|K|$ with this topology is called the *underlying space* of K or the *polytope* of K.

The ***dimension*** of a simplicial complex is the maximal dimension of a simplex in the complex.

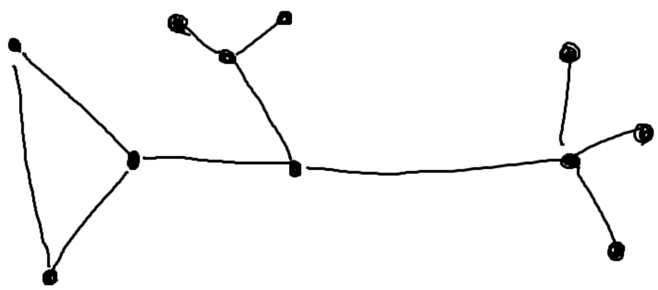
Clarification of the *face* concept



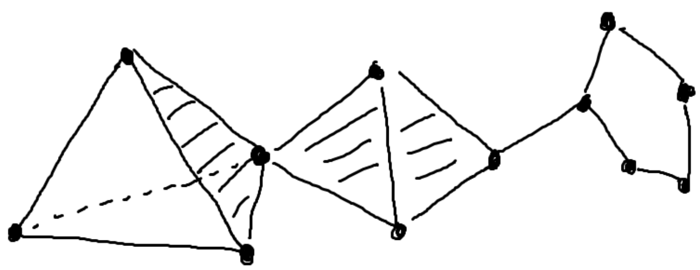
The ***boundary*** of an n-simplex Δ^n is defined as the union of all its faces of dimension smaller than n. It turns out that $\partial\Delta^n$ is homeomorphic to the $(n-1)$-sphere $S^{n-1} = \{x \in R^n \mid ||x|| = 1\}$:



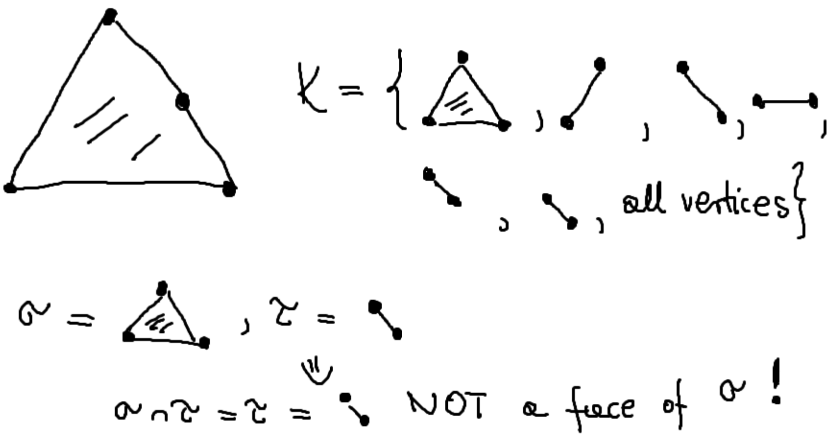
Example (graph)



Example ("generic" higher-dimensional simplicial complex)



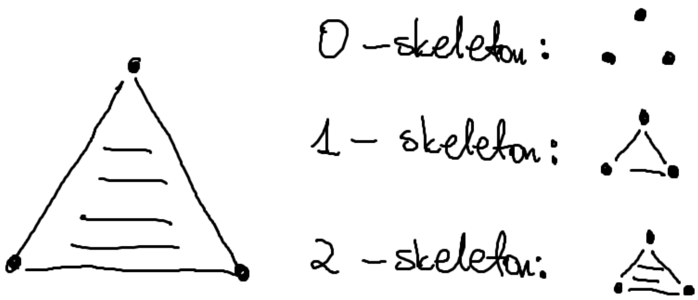
Example (not a simplicial complex)



1.3.3 Skeleta

The n-***skeleton*** $K^{(n)}$ of a simplicial complex K is defined by

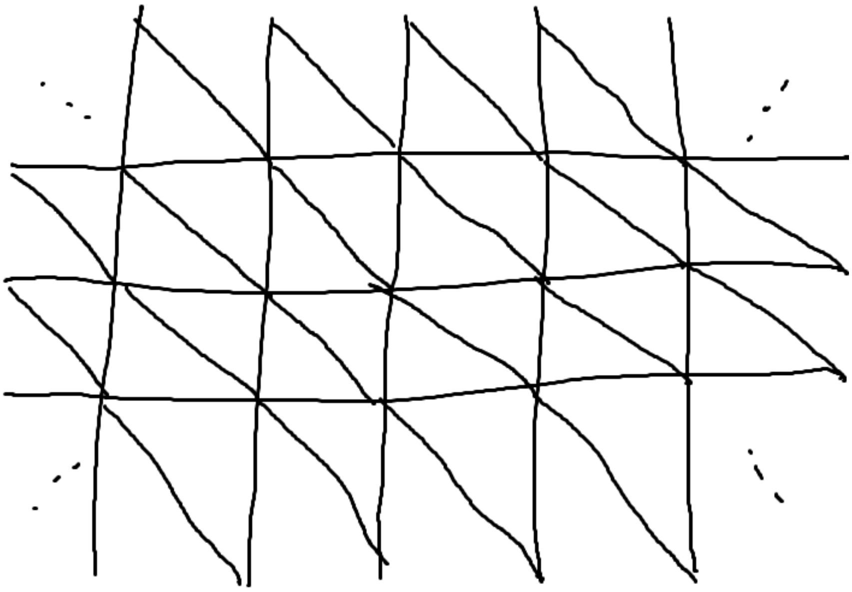
$$K^{(n)} = \{\sigma \in K \colon \dim \sigma \leq n\}. \tag{4}$$



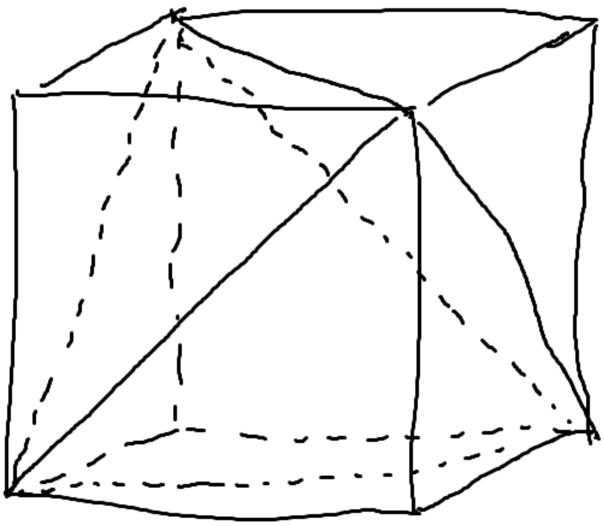
1.3.4 Triangulations

A ***triangulation*** of a topological space X is a homeomorphism $|K| \xrightarrow{\cong} X$ with K being a simplicial complex. We call X in this case a ***triangulable space***.

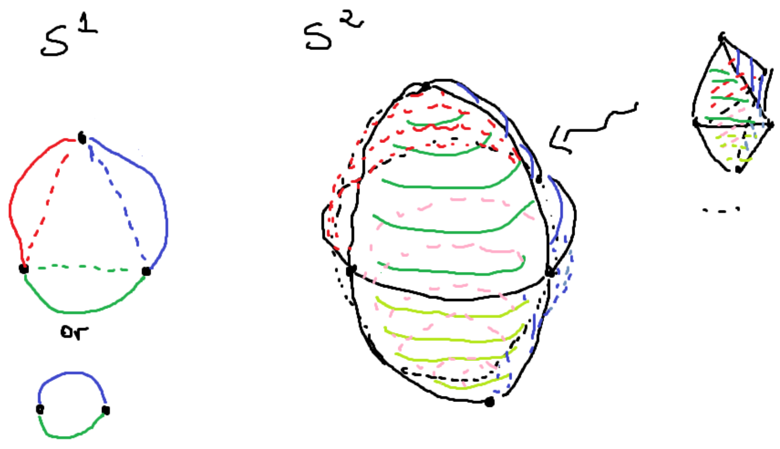
Example (a triangulation of R^2)



Example (a triangulation of a cube)

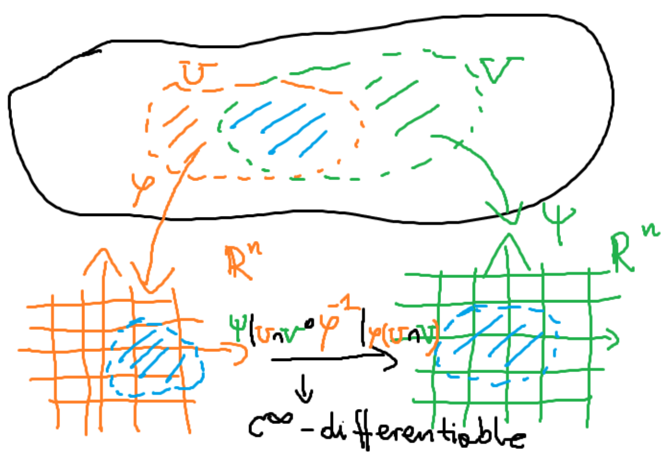


Example (triangulations of spheres)



Triangulability of manifolds

Smooth manifolds



Any smooth manifold is triangulable. However, there exist topological manifolds that do not admit a triangulation. The classical result states that up to dimension 3, all topological manifolds are polytopes, due to T. Radó (for surfaces – 2-manifolds) and E. Moise (3-manifolds).

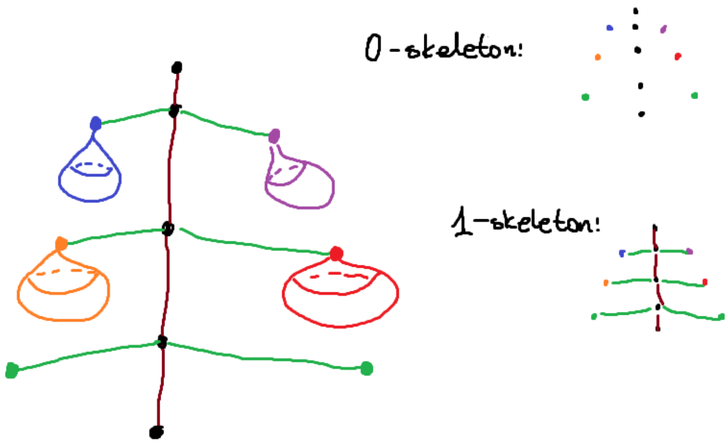
1.3.5 CW-complexes

A CW-complex X has an inductive definition as follows (we refer the reader to, e.g., Hatcher, p. 5):

- One starts with a discrete topological space X^0 and declares each point of this space to be a 0-cell of X.
- For $n \geq 1$, there are given maps $f_{n,\alpha} \colon D^n \to X^{n-1}$, called the *attaching maps*, defining X^n to be the quotient space of the disjoint union of n-disks D^n for each index α and X^{n-1} by the equivalence relation given by the identifications $x \sim f_{n,\alpha}(x)$.
- $X = \cup_{n \geq 0} X^n$, and one declares $A \subseteq X$ to be open if and only if $A \cap X^n$ is open in X^n for every $n \geq 0$.

In the case there are no attaching maps for each $m > n$, one has $X = X^n$. One calls the subspaces X^k the *k-skeleta* of X. Obviously, $X^0 \subseteq X^1 \subseteq X^2 \subseteq \cdots \subseteq X$. In this case, one calls the maximal n for which there exists an n-cell the *dimension* of X.

A CW-complex is called *finite* if it consists of finitely many cells.



CW-complexes are generalizations of simplicial complexes. Conversely, in a certain sense, every CW-complex X can be thought of as a simplicial complex (if X is finite, then the simplicial complex can be chosen to have the same dimension as X). This is indeed the case when considering topological spaces up to *homotopy equivalence*. This conclusion follows from the Simplicial Approximation Theorem — we refer the reader to HATCHER for details. It is convenient to be aware of both simplicial and CW-structures of a topological space, as either of them allows one to tackle specific aspects — often non-overlapping for the CW and simplicial cases — more efficiently.

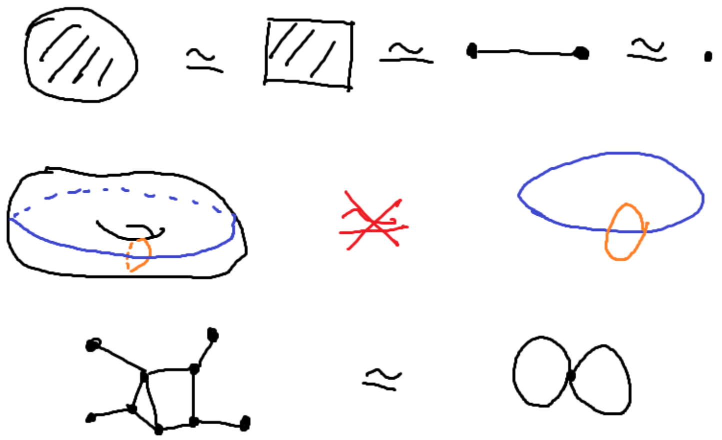
1.4 Basic invariants

1.4.1 Homotopy equivalence relation

A ***homotopy*** between maps $f\colon X \to Y$ and $g\colon X \to Y$ is a continuous map $H\colon X \times I \to Y$, $(x,t) \mapsto h_t(x)$, such that $h_0 = f$ and $h_1 = g$. In this case, we call f and g ***homotopic*** maps and denote this by $f \simeq g$.

Two topological spaces X and Y are called ***homotopy*** equivalent if there exist maps $f\colon X \to Y$ and $g\colon Y \to X$ such that $f \circ g \simeq \mathrm{id}_X$ and $g \circ f \simeq \mathrm{id}_Y$. The relation of being homotopy equivalent is an equivalence relation for topological spaces. One often considers topological spaces not up to homeomorphism but rather up to homotopy. One has to bear in mind that some information is lost, such as the dimension of the space. On the other hand, many tools from algebra, which we shall learn later, are actually sensitive only to the homotopy type of topological spaces.

Examples



1.4.2 Fundamental group

Now that we have learned about homotopy, we are ready to introduce one of the basic *algebraic invariants* of topological spaces, the ***fundamental group***. Denoted by $\pi_1(X)$ and defined for a (path)-connected topological space X, it is the set of homotopy classes of *loops* at a fixed $x_0 \in X$. Two loops $f\colon S^1 \to X$ and $g\colon S^1 \to X$ are defined to be homotopic if there exists a continuous map $H\colon S^1 \times I \to X$, $(\omega, t) \mapsto h_t(\omega)$, such that $f = h_0$, $g = h_1$, and $h_t(\omega_0) = x_0$ for all $t \in I$ and some fixed $\omega_0 \in S^1$.

It is convenient to look at S^1 in the definition above as the interval I with identified endpoints. In this way, one can reformulate the definition of the fundamental group as the set of homotopy classes of maps $f\colon I \to X$ satisfying $f(0) = f(1) = x_0$ for some fixed $x_0 \in X$. The homotopy between such maps f and g is a continuous map $H\colon I \times I \to X$, $(u,t) \mapsto h_t(u)$, such that $f = h_0$, $g = h_1$, and $h_t(0) = h_t(1) = x_0$ for all $t \in I$.

Looking at the definitions above, we realize that we have so far defined $\pi_1(X)$ as a set only. The group operation is given by the ***concatenation of loops***. To define it, we shall use the second version of the definition of the fundamental group.

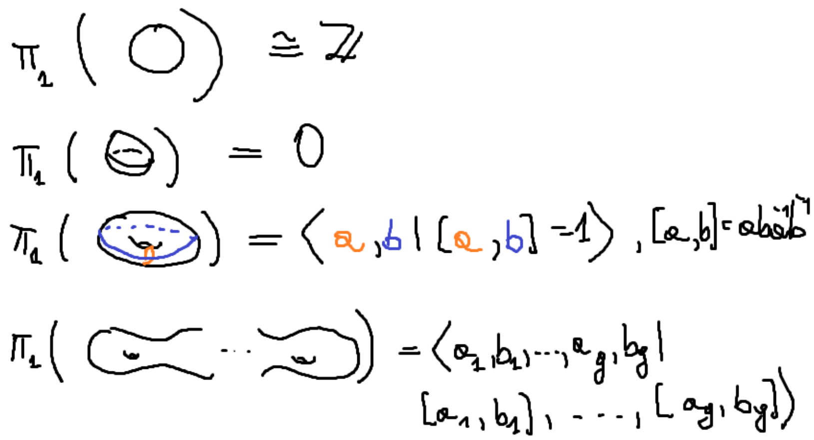
Let $f\colon I \to X$ and $g\colon I \to X$ be two maps such that $f(0) = f(1) = g(0) = g(1) = x_0$. The concatenation of f and g is the map $f * g\colon I \to X$ defined as follows:

$$f * g(t) = \begin{cases} f(2t) & \text{for } t \leq 1/2, \\ g(2t-1) & \text{for } t \geq 1/2. \end{cases} \tag{5}$$

The concatenation of loops descends to $\pi_1(X)$, i.e., the operation $*\colon \pi_1(X) \times \pi_1(X) \to \pi_1(X)$, $([f],[g]) \mapsto [f * g]$, is well-defined. Moreover, it satisfies the group axioms. The identity in $\pi_1(X)$ is the homotopy class of constant maps at x_0, and the inverse of an element $[f] \in \pi_1(X)$ is $[\overline{f}]$, where $\overline{f}$ denotes the *reverse* loop of f, i.e., $\overline{f}(t) = f(1-t)$.

It should be noted that the choice of $x_0 \in X$ does not affect $\pi_1(X)$.

Examples



1.4.3 Euler characteristic

Let X be a simplicial complex of dimension n with a finite number of faces: k_0 0-dimensional, k_1 1-dimensional, ..., k_n n-dimensional. Then the Euler characteristic of X, denoted by $\chi(X)$, is defined as

$$\chi(X) = k_0 - k_1 + \ldots + (-1)^n k_n. \tag{6}$$

The same definition can be introduced for a finite CW-complex X (of dimension n) with k_i i-cells:

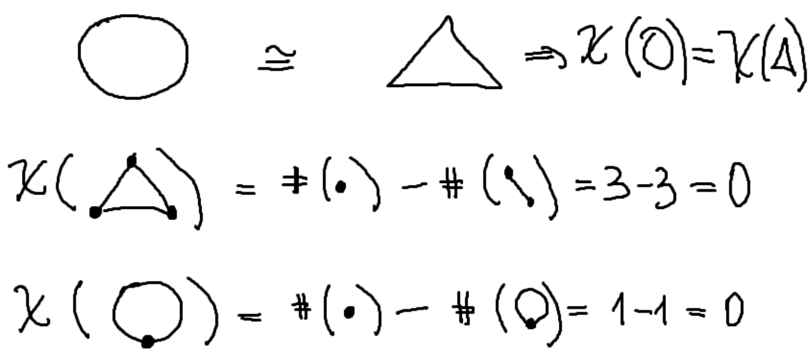
$$\chi(X) = k_0 - k_1 + \ldots + (-1)^n k_n. \tag{7}$$

If a topological space is endowed with both a CW-structure and a simplicial structure, then it does not matter from which of these structures we compute the Euler characteristic.

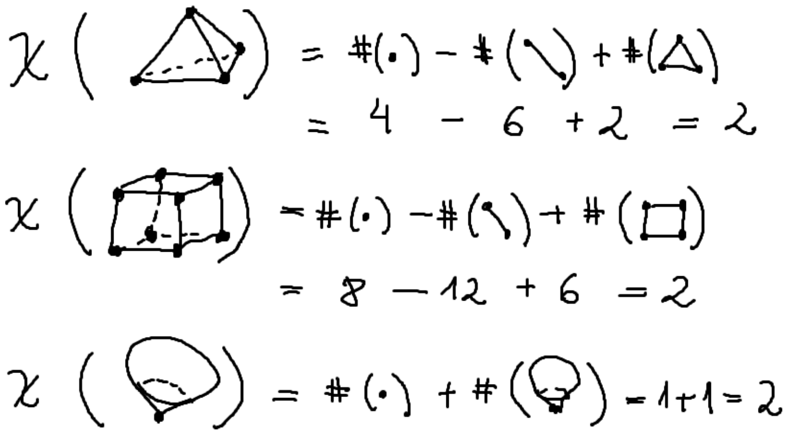
Actually, one can define the Euler characteristic for any topological space X, and this definition — equivalent to the one above for CW-complexes — is sensitive only to the homotopy type of X. In order to define $\chi(X)$ in such generality, we first need to introduce the concept of ***homology***. We shall do this in the next lecture.

Example (Euler characteristic of the sphere S^n)

The case $n = 1$



The case $n = 2$ All surfaces of connected convex (this, actually, is not that important) polyhedra are homeomorphic to the sphere S^2. Thus, they share the same Euler characteristic. For such surfaces, this recovers the well-known formula $V - E + F = 2$, where V, E, and F denote the numbers of vertices, edges, and faces of the given polyhedron.



The general case We can easily generalize the formula for $\chi(S^n)$ using the very simple CW-decomposition of S^n, already presented for $n = 1$ and $n = 2$. This decomposition consists of only one vertex and one n-dimensional cell, obtained by attaching the n-disk via the constant map onto that vertex. Thus,

$$\chi(S^n) = 1 + (-1)^n. \tag{8}$$

1.5 Programming tasks

Programming Task 1: Verify Metric Axioms on a Finite Set (Python)

Goal: Determine whether a given distance function on a finite set satisfies the axioms of a metric: non-negativity, identity of indiscernibles, symmetry, and triangle inequality.

Input:

- An integer `n` — number of points in the set $X = \{0, 1, \ldots, n-1\}$.
- A real-valued matrix `D` of size `n × n`, where `D[i][j]` represents $d(i, j)$.

Output:

- The string `"METRIC"` if all axioms are satisfied.
- Otherwise the string `"NOT A METRIC"` and a list of violated axioms, chosen from:
  - `"nonnegativity"` if there exist `i, j` with $d(i, j) < 0$,
  - `"identity"` if there exist `i ≠ j` with $d(i, j) = 0$ or some `i` with $d(i, i) \neq 0$,
  - `"symmetry"` if there exist `i, j` with $d(i, j) \neq d(j, i)$,
  - `"triangle"` if there exist `i, j, k` with $d(i, k) > d(i, j) + d(j, k)$.